

JavaScript Symbol Datatype - Notes

1. What is Symbol?

Symbol is a primitive datatype used to create unique values.

2. Syntax

```
let a = Symbol();  
let b = Symbol('id');
```

3. Typeof

`typeof Symbol()` -> symbol

4. Unique Nature

`Symbol('x') === Symbol('x')` -> false
Every new Symbol is unique.

5. Object Keys

Symbols are often used as object property keys to avoid name conflicts.

6. Hidden Properties

Symbol keys are skipped in normal `for...in` loops.

7. Global Registry

`Symbol.for('token')` returns shared global symbol.

8. String Conversion

Use `sym.toString()` to convert symbol into text.

9. Boolean Conversion

`Boolean(Symbol())` -> true

10. Best Uses

Internal IDs, constants, hidden metadata, safe object keys.

11. Symbol vs Symbol.for()

Feature	Symbol()	Symbol.for()
Uniqueness	Always new unique symbol	Shared by key
Comparison	Different each time	Same key gives same symbol
Use Case	Private keys	Global constants